

ECAM procedures from the A380 FCOM

ABN PROC (not-sensed) procedures and the sensed alerts that have FCOM action lines

Branch	feature/a380/fws/abn-proc-procedures
Aircraft	A380X
Commits	00b3a980d feat(a380x/fws): ECAM procedures from the A380 FCOM for the ABN PROC and the sensed alerts
Base	pr/10-a380x-mfd-fcom-pages (first built on the old fork master 280800cf7)
Status	Merged in develop. Built, tsc/lint clean, consistency-checked; deployed, in-sim test pending.

Problem

Thirteen ABN PROC procedures were marked WIP with no lines, and a not-sensed procedure is only shown on the EWD if it has an entry in `FwsAbnormalNonSensed`: activating them showed nothing. 102 alerts wired in the FWS had no items.

Specification

ABN PROC (A380 FCOM PRO-ABN-ECAM-20; texts from PRO-ABN-ECAM-10)

Procedure	Menu	Notes
MISC DITCHING	first page	FCOM text incl. approach, 2000 / 500 ft, after ditching
MISC FORCED LANDING	first page	engines running case
MISC CKPT WINDOW CRACKED / ELEC ARCING	MISCELLANEOUS	external / internal layer, AICU 1 / 2
MISC OIS FAULT	MISCELLANEOUS	OIT / laptop restart
FUEL LEAK	FUEL	the FUEL LEAK DETECTED procedure; LAND ANSA
FUEL MAN BALANCING PROCEDURE	FUEL	DELAY T.O on ground; feed / inner / mid tank balancing in flight
L/G LDG WITH ABNORM ... (5 procedures)	L/G	WHEEL SD page
NAV IR ALIGNMENT IN ATT MODE	NAV	
ENG HI VIBRATIONS (new)	ENG	in the FCOM ENG list, was missing

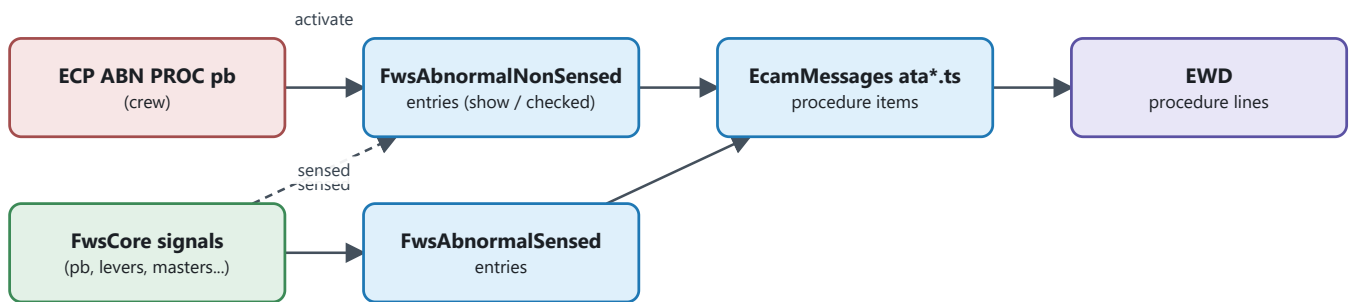
Sensed alerts (A380 FCOM PRO-ABN-ECAM-10)

- Of the 102 wired alerts without items, about 85 are "Crew awareness" in the FCOM (title only, plus STATUS / redundancy lines): their empty item list is correct. 4 are not in the 2011 FCOM (HOT AIR 1(2) OFF / VLV OPEN, T.O V1/VR/V2 DISAGREE, OVERSPEED LOAD ANALYSIS REQUIRED) and stay empty.
- Filled: COND HOT AIR 1(2) FAULT, COND TEMP CTL FAULT, ENG 1(2)(3)(4) FAIL (in flight / on ground, engine number in the lines, sensed thrust lever / master / fire pb / agent), ENG ALL ENGINES FAILURE (FCOM ALL ENG FLAME OUT, fuel remaining / no fuel, forced landing and ditching; nesting capped at three levels for the EWD width).

FCOM conventions applied

- Upper case "IF ...:" is an ECAM condition line; lower case "If ...:" is a display condition (it selects which lines are shown, in `whichItemsToShow`).
- Headers like "AT 2000 FT AGL :-" are centred sub-headlines; lines are at most 41 characters (EWD width).
- A line is sensed only where the FWS has the signal (seat belts, masters, fire pb, agents, park brake, RAM AIR, L/G lever, ground spoilers, GPWS / TERR off, A/THR, thrust levers, APU master / bleed, EVAC, batteries).

■ Component added or changed by the feature
 ■ MSFS / simulator
 ■ Aircraft system (systems host, FMS, FWS)
 ■ Display / page
■ Data, LVar, storage
■ Crew, flypad, external program



Architecture

Verification

- Every item list vs its show / checked arrays checked by script over all 611 arrays of the two FWS files: consistent (one harmless pre-existing extra in VOLCANIC ASH).
- systems host, EWD, PFD, SD build; tsc systems-host clean; ESLint and Prettier clean.

Limitations

- Deferred procedures, limitations and STATUS of the new procedures are not written yet.
- FLUCTUATING VERTICAL SPEED stays WIP (not in the 2011 FCOM).
- Where the FCOM lets the ECAM pick a side (e.g. L/G affected side) and FBW cannot sense it, a condition line is used.

PR description (from the PR draft)

From `pr-drafts/17-ecam-procedures.md`, section "Summary of Changes" onwards.

Summary of Changes

A not-sensed procedure is only displayed with its entry in `FwsAbnormalNonSensed`: the ABN PROC procedures marked WIP had neither lines nor entry and showed nothing when activated. The procedures now carry the A380 FCOM text and their FWS entries (sensed lines where the FWS has the signal).

ABN PROC (FCOM PRO-ABN-ECAM-20; texts PRO-ABN-ECAM-10): MISC DITCHING, FORCED LANDING, CKPT WINDOW CRACKED, CKPT WINDOW ELEC ARCING, OIS FAULT; FUEL LEAK (the FUEL LEAK DETECTED procedure, LAND ANSA), MAN BALANCING PROCEDURE; L/G LDG WITH ABNORM NOSE L/G, WITH 2 ABNORM L/Gs ON SAME SIDE, WITH 1 ABNORM WING OR BODY L/G, WITH 2 ABNORM BODY L/Gs, WITH 2 ABNORM WING L/Gs (WHEEL page); NAV IR ALIGNMENT IN ATT MODE; new ENG HI VIBRATIONS (in the FCOM ENG list, was missing).

Sensed alerts (FCOM PRO-ABN-ECAM-10): of the 102 wired alerts without lines, about 85 are "Crew awareness" in the FCOM (title only): left empty. Filled: COND HOT AIR 1(2) FAULT, COND TEMP CTL FAULT, ENG 1(2)(3)(4) FAIL (in flight / on ground, the engine number in the lines, thrust lever / master / fire pb / agent sensed, L/G UP after lift-off), ENG ALL ENGINES FAILURE (the FCOM ALL ENG FLAME OUT: fuel remaining / no fuel, relight, forced landing and ditching).

FCOM conventions: upper case "IF ...:" lines are ECAM condition lines; lower case "If ...:" only select the lines shown. Headers ("AT 2000 FT AGL :") are centred sub-headlines; lines fit the 41-character EWD width (nesting capped at three levels).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: each ABN PROC procedure on the EWD, ENG 1 FAIL in flight and on ground, ALL ENGINES FAILURE.

References

- A380 FCOM PRO-ABN-ECAM-10 (procedures), PRO-ABN-ECAM-20 (not-sensed ABN PROC list).

Additional context

Not done: the deferred procedures, limitations and STATUS of the new procedures; FLUCTUATING VERTICAL SPEED stays WIP (not in the 2011 FCOM). HOT AIR 1(2) OFF / VLV OPEN, T.O V1/VR/V2 DISAGREE and OVERSPEED LOAD ANALYSIS REQUIRED are not in this FCOM and stay without lines.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. ECP ABN PROC: open each procedure of the list, activate it: the lines show; sensed lines tick (e.g. SIGNS, ALL ENG MASTERS, PARK BRK).
2. Fail an engine in flight: ENG x FAIL with AUTO RELIGHT IN PROGRESS, THR LEVER x / ENG x MASTER; on ground: the ground lines.
3. Fail all engines in flight: ALL ENGINES FAILURE, LAND ASAP, the full procedure scrolls.
4. HOT AIR 1 pb OFF: COND HOT AIR 1 FAULT with HOT AIR 1 ON.